GEONOS INFRASTRUCTURE · TECHNICAL & COMMERCIAL REPORT

Climate-conditioned infrastructure decisions

A full energy, cost, peak-load, and resilience report connecting a real Hunts Point cold-chain benchmark with a hypothetical 24 MW data center.

CORE QUESTION

Can climate risk become a present design variable—not a future footnote?

GEONOS joins climate projections, facility load, energy price, peak demand, and battery duration in one traceable decision model.

01	24 MW	10	4
real public project	hypothetical new build	traceable evidence sources	provenance classes

Prepared by GEONOS | New York City | July 2026

01 · EXECUTIVE DECISION

The product sells design decisions—not a forecast

RECOMMENDATION

Use Hunts Point as the first validation market and data centers as the second module.

Both assets remove heat continuously, but cold storage is governed by product thermal mass and door operations, while data centers are governed by IT load, PUE, and power continuity. Keep one climate/grid core with separate vertical physics.

What the report establishes

- Public Hunts Point facts: approximately 1,000,000 ft^2 total, more than 800,000 ft^2 refrigerated, over 2.5 billion lb/year of produce, 25% of NYC fresh produce, and a goal to eliminate 1,000 diesel refrigerated trailers.
- The Hunts Point planning case yields 106.4 GWh of 2050s climate-adjusted electricity, a $22.50M 2025 price benchmark, and a 11.1 MW mitigated peak.
- Hypothetical Project Atlas—24 MW IT, 85% utilization, PUE 1.23—yields 229.3 GWh, $48.25M, and a 21.9 MW mitigated peak.
- Neither result is measured facility performance. Bankability requires interval load, product-temperature or IT-load data, equipment curves, an applicable tariff, and utility/interconnection terms.

The four decisions GEONOS sells

WHEN	HOW MUCH	WHAT	PROOF
when the peak arrives	climate cost and capacity	which design mix mitigates it	which data makes it bankable

02 · EVIDENCE ARCHITECTURE

Four provenance classes prevent false precision

Class	Definition	Example in this report
PUBLIC	Published project, climate, price, or emissions fact	Hunts Point area; NPCC CDD; NYSERDA price
ACTUAL DATA	Observed weather or market input	NOAA temperature normals; NYISO inputs to research
RESEARCH	Regression or simulation output from actual inputs	+100 MW mean fitted price increment of $0.679/MWh
MODELED	Editable planning input where site data are absent	Cold-storage EUI 100; Atlas PUE 1.23; BESS sizes
HYPOTHETICAL	Concept with no implied parcel, permit, or grid approval	Project Atlas in Long Island City

Calculation chain

Output	Simplified equation	Interpretation
Data-center energy	IT MW × utilization × 8,760 × PUE	Includes modeled facility overhead

Output	Simplified equation	Interpretation
Cold-storage energy	Area × site EUI × system factor	Needs interval and end-use validation
Climate factor	CDD growth × asset cooling sensitivity	Translates NPCC CDD into a design stress
Monthly cost	Monthly MWh × NY commercial ¢/kWh	All-in benchmark, not a site tariff
Battery autonomy	MW × duration × 90% / critical MW	Excludes generator and UPS sequencing

The equations match the live web model. Every planning input is editable and all outputs update together.

03 · REAL DATA STACK

New York climate, price, and emissions baseline

1,156	1,919	14–19 in	537 lb/MWh
1981–2010 CDD	2050s central CDD	2050s sea-level rise	2024 NY emissions

Monthly inputs

Month	NOAA LGA normal °F	2025 NY commercial ¢/kWh	Applied 2050s °F
Jan	34.4	20.4	38.2
Feb	36.3	20.9	40.1
Mar	43.1	20.3	46.9
Apr	53.6	19.4	57.4

Month	NOAA LGA normal °F	2025 NY commercial ¢/kWh	Applied 2050s °F
May	63.7	19.5	67.5
Jun	73.4	22.0	77.2
Jul	79.2	23.1	83.0
Aug	77.7	22.4	81.5
Sep	70.8	22.7	74.6
Oct	59.6	20.8	63.4
Nov	49.1	19.8	52.9
Dec	40.0	20.6	43.8

Temperature: NOAA 1991–2020 Climate Normals, LaGuardia USW00014732. Price: NYSERDA/EIA 2025 statewide commercial all-in average. The +3.8°F translation is a GEONOS scenario device calibrated to the central 2050s CDD value; it is not an official NPCC temperature projection.

04 · CASE 01 / PUBLIC PROJECT

Hunts Point: electrification creates a new design problem

1.0M ft²	800k+ ft²	2.5B lb/yr	25%
planned total area	refrigerated area	co-op produce volume	NYC fresh-produce share

The Hunts Point Produce Market redevelopment is a real public project. NYC materials describe an all-electric intermodal food-distribution center, a $405 million committed public funding package, expected construction beginning in late 2026, and air-quality benefits for nearly

13,000 nearby residents. Eliminating 1,000 stationary diesel refrigerated trailers moves the design problem toward grid capacity, electric peak, and outage resilience.

Boundary between public fact and planning assumption

Item	Value	Provenance	Meaning
Gross area	1,000,000 ft^2	PUBLIC	NYC-published project scale
Refrigerated area	>800,000 ft^2	PUBLIC	Cold-chain dominant program
Site EUI	100 $kWh/ft^2 \cdot yr$	MODELED	No public measured load; editable
Refrigeration	Transcritical CO_2	MODELED	Concept alternative, not final design
Battery	8 MW × 4 h	MODELED	Peak and critical-load test
Climate horizon	Central 2050s	PUBLIC + MODELED	NPCC central CDD applied

CORE RISK

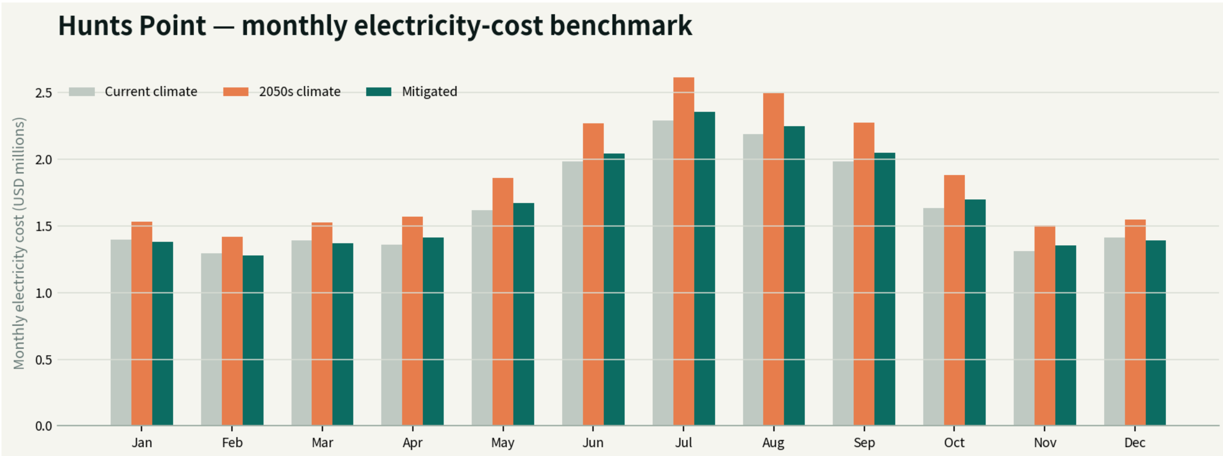
Electrified heat rejection joins operating cost, grid capacity, and food safety.

Door opening, inventory thermal mass, outdoor temperature, compressor control, defrost, trailer power, and emergency operation belong on one timeline.

05 · CASE 01 / MODEL OUTPUT

Hunts Point monthly energy and cost

106.4 GWh	$22.50M	$2.63M	11.1 MW
2050s climate energy	2025 price benchmark	climate cost premium	mitigated peak



Orange is the 2050s climate case. Green applies 9% efficiency and 12% flexible load. The battery demand-value proxy is separate and is not added to energy savings.

Metric	Current climate	2050s climate	Mitigated
Annual energy	94.0 GWh	106.4 GWh	95.8 GWh
Annual cost	$19.87M	$22.50M	$20.26M
Peak	15.8 MW	17.4 MW	11.1 MW

06 · CASE 01 / BANKABILITY

The first pilot sells a closed data loop—not a savings claim

DECISION

An 8 MW × 4 h battery does not pass a 12-hour food-safety test alone.

At a critical-load assumption of 40% of peak, modeled autonomy is 4.1 h and the remaining gap is 7.9 h. Re-size with thermal storage, temperature bands, generation, and inventory rotation.

90-day pilot

1. Days 0–30 / Baseline: synchronize 15-minute interval electricity, outdoor weather, room temperature, doors, compressor/fan/defrost state, and inventory throughput.
2. Days 31–60 / Identify: estimate load sensitivity, thermal inertia, infiltration, controllable load, and food-safety boundaries.
3. Days 61–90 / Validate: test pre-cooling, setpoint bands, defrost scheduling, and battery dispatch in shadow mode and controlled events.

Success gates

Gate	Required result	Commercial meaning
Measurement	≥95% data coverage; <1 interval alignment error	Auditable baseline
Prediction	Peak timing and load error within pre-agreed range	Design confidence
Safety	No product-temperature or HACCP violation	Operator acceptance
Economics	NPV range including tariff, incentives, and capex	Bankable decision

07 · CASE 02 / HYPOTHETICAL NEW BUILD

Project Atlas: a 24 MW Queens data center

HYPOTHETICAL

No parcel, permit, or interconnection is implied.

Long Island City is an analytical coordinate for applying the regional data stack. Atlas is a concept facility that demonstrates the decisions GEONOS can support during design.

24 MW	85%	1.23	12 MW × 4 h
IT design capacity	IT utilization	liquid-cooling PUE	battery concept

Design brief

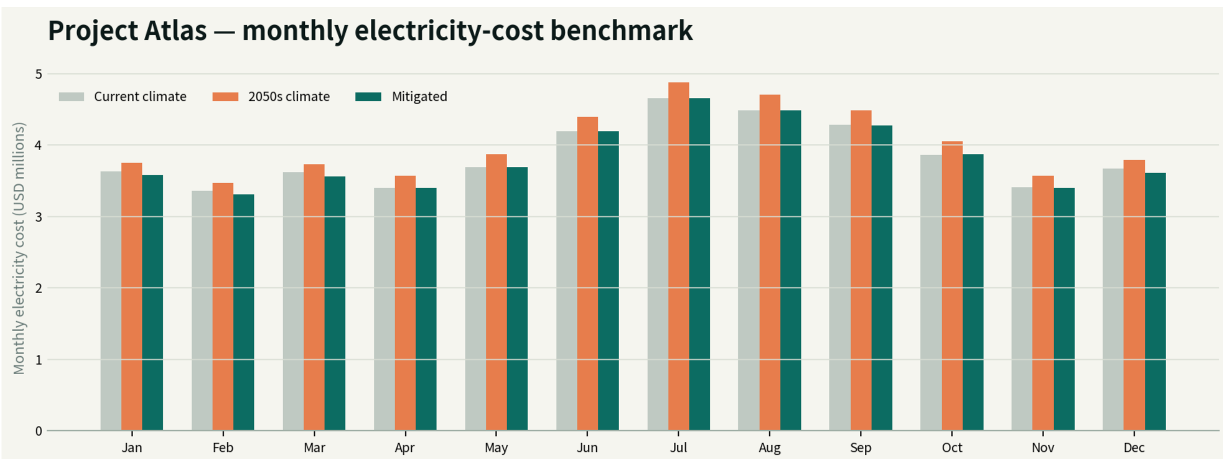
Design area	Base concept	GEONOS validation question
IT load	24 MW; 85% utilization	What is the hourly workload and growth curve?
Heat rejection	Direct liquid; PUE 1.23	How does PUE move with weather and part load?
Power	Con Edison / NYISO Zone J	What service, ramp, curtailment, and upgrade terms apply?
Resilience	12 MW × 4 h BESS	How do UPS, generators, and critical loads sequence?
Heat reuse	Space reserved	Does nearby demand match temperature and season?
Water	WUE tracked	What is the water/energy/noise trade-off by option?

LBNL reports average hyperscale/colocation PUE below 1.4 by 2023 and projects a 2028 average PUE range of 1.15–1.35. Atlas 1.23 is a concept assumption within that range.

08 · CASE 02 / MODEL OUTPUT

Project Atlas monthly energy and cost

229.3 GWh	$48.25M	$2.00M	21.9 MW
2050s climate energy	2025 price benchmark	climate cost premium	mitigated peak



The cost is not an Atlas bill. It is a planning benchmark applying NYSERDA's 2025 statewide commercial all-in monthly average to the modeled load.

Metric	Current climate	2050s climate	Mitigated
Annual energy	219.8 GWh	229.3 GWh	218.7 GWh
Annual cost	$46.25M	$48.25M	$46.01M
Peak	29.5 MW	32.6 MW	21.9 MW

09 · CASE 02 / GRID & RESILIENCE

Keep the 24 MW project separate from the 100 MW system benchmark

GEONOS RESEARCH

$0.679/MWh mean fitted price increment at +100 MW

A GEONOS research export using actual NYISO inputs produced $0.595M/year in modeled system exposure, 1.39% of baseline wholesale cost, and 0.365% of hours outside model support under 2025 conditions.

Project Atlas is 24 MW—below the research workflow's 100 MW minimum. The result is therefore not linearly scaled or claimed as a project-price forecast. It is a directional regional stress benchmark showing why aggregated load and targeted peak avoidance matter.

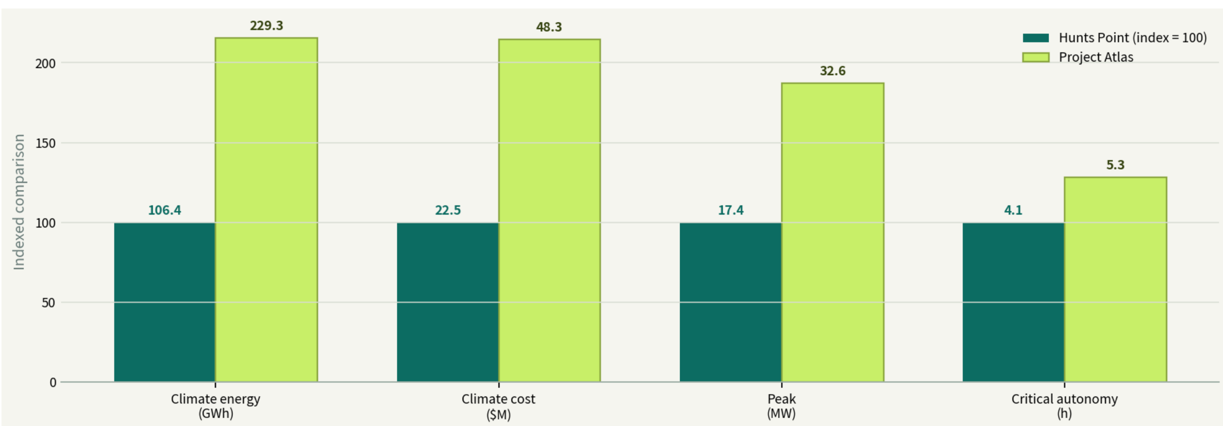
Constant added load	Mean fitted increment	Annual modeled exposure	Interpretation
100 MW	$0.679/MWh	$0.595M	Minimum research benchmark
250 MW	$1.704/MWh	$3.732M	Cluster-growth scenario
500 MW	$3.432/MWh	$15.029M	High-density stress
1,000 MW	$6.959/MWh	$60.956M	System stress

Atlas resilience

5.3 h	1.3 h	2.7 h	$2.28M
25% critical-load bridge	full-load bridge	gap to 8-hour target	annual demand proxy

10 · CROSS-CASE COMPARISON

One engine, different physics and buyers



Values printed over the bars retain their native units; bar heights index Hunts Point to 100. This visual compares model structure, not asset superiority.

Dimension	Cold storage	Data center
Heat source	Weather, product, infiltration, doors, defrost	Nearly all IT electricity becomes heat
Flexibility	Thermal mass, pre-cooling, defrost shifting	Workload, cooling, UPS/BESS
Hard constraint	Product temperature, food safety, shipping	Uptime, latency, power quality
First buyer	Owner, operator, refrigeration engineer	Developer, operator, utility/interconnect
Shared outputs	Load, peak, cost, climate, autonomy, next data	Load, peak, cost, climate, autonomy, next data

11 · SENSITIVITY

Show decision ranges before precise-looking point estimates

Case	Sensitivity	Low	Base	High	Evidence needed
Hunts	EUI kWh/ft²·yr	70	100	130	12-month interval + end use
Hunts	Efficiency	4%	9%	14%	Controls tests and M&V
Hunts	Flexible load	5%	12%	20%	Product temperature, doors, inventory
Atlas	IT utilization	70%	85%	95%	Hourly workload curve
Atlas	PUE	1.15	1.23	1.35	Part-load performance curves
Atlas	BESS duration	2h	4h	8h	UPS/generator sequence
Shared	Price	Contract/ hedge	2025 NY avg.	High-cost tariff	Utility service study
Shared	Climate	Current	2050s central	2050s 75th	Site downscaling

PRODUCT PRINCIPLE

GEONOS provides a decision boundary—not a single magic number.

It identifies which input can reverse the decision and which next evidence reduces uncertainty most.

LEVEL 0 **LEVEL 1** **LEVEL 2**

public-data screen	data-room model	validated pilot

12 · COMMERCIAL PRODUCT

From interactive demo to recurring revenue

Product	Buying moment	Deliverable	Revenue model
Climate Screen	Site selection / early development	Climate-grid-price screen, ranges, go/no-go	$15k–30k per study
Design Underwrite	Concept / schematic design	Monthly load, options, peak/resilience, data gap	$60k–150k per project
Operational Twin	After commissioning	Forecast, dispatch advice, M&V, reporting	$5k–15k/site/month
Portfolio API	Multi-asset management	Risk scores, capex priority, audit trail	$100k+ enterprise ARR

Defensibility

- The moat is not public data alone; it is the joined facility operating data, vertical physics, validation record, and decision audit trail.
- Cold storage and data centers should not collapse into one generic EUI calculator. They share a climate/grid core but retain vertical-specific modules.
- Recurring revenue requires a prediction–operation–M&V loop, not one-off report sales.

Economic buyer

DEVELOPER	OPERATOR	LENDER	UTILITY
power, site, capex	cost, uptime, quality	stress and covenants	interconnect and flexibility

13 · EXECUTION ROADMAP

Twelve months: prove one site, then replicate the second vertical

Period	Objective	GEONOS work	Completion gate
0–4 weeks	Demo → selling tool	Lock web model, ledger, input schema, and proposal	Five-user usability review
Months 1–3	Cold-storage design partner	NDA, data request, baseline, walkthrough	12 months or representative data
Months 3–6	Shadow pilot	Load forecast, event replay, control advice	Pre-agreed accuracy and safety gates
Months 6–9	Validate and commercialize	M&V, capex options, tariff/DR, lender pack	Paid renewal or second site
Months 9–12	Data-center module	IT profile, PUE curve, UPS/BESS, interconnect	One developer design review

What to do now

4. Approach Hunts Point stakeholders with a public-data design screen that makes no claim about actual project performance.

5. Interview ten cold-storage operators and score access to interval electricity, temperature, door, and controls data.
6. Put data rights, anonymization, test approval, M&V, and publication rights in the pilot LOI.
7. Use Project Atlas as proof of capability without implying a real parcel or interconnection.

14 · LIMITATIONS, DATA REQUEST & SOURCES

What makes the model bankable

Minimum data package

Shared	Cold-storage additions	Data-center additions
15-minute electricity, tariff, contract demand	Room temperature/humidity/setpoints	IT load, UPS efficiency, PUE
One-line, equipment list, maintenance	Doors, defrost, compressor stage	Rack density, workload, SLA
Weather, outage, and event logs	Inventory, product type, throughput	Cooling loops, water, generators
Utility and interconnection files	HACCP/product-temperature limits	Commissioning and failure sequence

Material limitations

This is a planning model. Hunts Point electricity use, refrigeration design, tariff, and battery are not public and are modeled. Project Atlas is entirely hypothetical. NYSERDA prices are statewide averages; the Con Edison value is an illustrative delivery-demand proxy; the EIA emissions factor is annual. Citywide sea-level-rise ranges do not replace a parcel flood study. Economics omit capex, financing, degradation, incentives, tax, and outage value.

Primary sources

1. NYC Mayor's Office / NYCEDC — Hunts Point Produce Market redevelopment LINK ↗
2. NYC Mayor's Office / NYCEDC — Historic Hunts Point redevelopment agreement LINK ↗
3. NOAA NCEI — 1991–2020 U.S. Climate Normals LINK ↗

4. NYC Panel on Climate Change / NYC Open Data — NYC Climate Projections: Extreme Events and Sea Level Rise LINK ↗
5. NYSERDA / U.S. EIA — Monthly Average Retail Price of Electricity — Commercial LINK ↗
6. Con Edison — SC 9 Rate II phase-in delivery charges LINK ↗
7. U.S. Energy Information Administration — New York Electricity Profile 2024 LINK ↗
8. Lawrence Berkeley National Laboratory — 2024 United States Data Center Energy Usage Report LINK ↗
9. Lawrence Berkeley National Laboratory — Industrial refrigerated warehouse demand-response study LINK ↗
10. GEONOS research workflow — NYISO added-load price-response export LINK ↗

Full URLs and usage notes also appear in the website's Source & assumption ledger. Model/report version: GEONOS Infrastructure v1.0, 2026-07-27.